

Odds Factory: Advanced Market Engine & AI Optimization Logic

1. Executive Summary

This document expands the Odds Factory backend architecture to handle deep derivative markets across 11 sports. Furthermore, it outlines the implementation of an AI-driven **Continuous Learning Engine** and a **Dynamic Market Parser**, allowing the backend to mathematically process, optimize, and edit bets for markets that are not explicitly defined in the core statistical logic.

2. Advanced Sport-Specific Market Expansions

Beyond the primary Winner and Totals markets, the Odds Factory engine must deploy secondary statistical models to accurately price and optimize derivative markets.

2.1. Association Football (Soccer)

- **Both Teams to Score (BTTS):** Derived from the Bivariate Poisson matrix by summing all probabilities where both Home > 0 and Away > 0 .
- **Corners (Total, Match Winner, Asian Lines):** Modeled using an independent Negative Binomial distribution. Unlike goals, corners occur in clusters (a deflected shot often leads to consecutive corners).
- **Cards / Booking Points:** Modeled using a state-dependent Poisson distribution. The baseline rate changes based on the referee's historical strictness and match state (e.g., a team losing 1-0 in the 80th minute commits more fouls).
- **Goalscorers (Anytime, First, Last):** Modeled using an expected goals (xG) share metric multiplied by the team's total Poisson expected goals (λ).

2.2. Basketball

- **Quarter / Half Betting:** The Gaussian (Normal) model is divided by 4, but variance is increased. The 4th quarter requires an inflator due to intentional fouling and free-throw sequences extending totals.
- **Player Props Combos (Points + Rebounds + Assists - PRA):** Requires a Copula function to model the correlation between variables (e.g., a high number of rebounds often correlates positively with points via put-backs).
- **Race to X Points:** Modeled via a discrete-time random walk based on the respective offensive efficiency ratings of both teams.

2.3. Tennis

- **Total Aces / Double Faults:** Modeled via independent Poisson distributions derived from player serve speed and second-serve risk profiles.

- **Tiebreak in Match:** Calculated from the Markov chain. If p (Player A serve win %) and q (Player B serve win %) are both extremely high (e.g., >80%, typical of John Isner), the probability of the set score reaching 6-6 approaches certainty.
- **Next Game Winner (Live):** Purely isolates the server's baseline p value, adjusted dynamically by live momentum (break points saved/converted in previous games).

2.4. Ice Hockey

- **Shots on Goal (SOG):** Modeled via Gaussian distribution based on Corsi/Fenwick metrics (total shot attempts).
- **Powerplay Points:** Modeled conditionally. The engine calculates the probability of Team A committing a penalty, multiplied by Team B's historical powerplay conversion rate.
- **Period Betting:** High variance. The 3rd period carries a completely different mathematical profile due to Empty Net scenarios.

2.5. Baseball

- **Hits, Runs, Errors (HRE):** A composite market requiring three separate distributions (Poisson for Errors, Negative Binomial for Hits/Runs) combined via Monte Carlo simulation.
- **Pitcher Strikeouts (K's):** Modeled against the opposing lineup's historical swing-and-miss (Whiff) rate.
- **First Inning Winner / YRFI (Yes Run First Inning):** Highly dependent on the top 3 batters in the lineup facing the starting pitcher before fatigue sets in.

2.6. Volleyball & Table Tennis & Badminton (Rally-Point Sports)

- **Exact Game/Set Margin:** Derived from the Markov Chain limit. E.g., The probability of a 25-18 volleyball set versus a 25-23 set.
- **Extra Points (Deuce / Setting):** The engine calculates the precise probability of a 24-24 (Volleyball) or 10-10 (Table Tennis) tie, which triggers win-by-two logic, altering total points models dynamically.

2.7. Cricket

- **Fall of Next Wicket:** Modeled using a Weibull survival distribution (commonly used in engineering to predict time-to-failure), predicting how many runs will be scored before the current partnership is broken.
- **Player Boundaries (Over/Under 4s and 6s):** Modeled using the batsman's historical boundary percentage applied to their projected balls faced.

2.8. Darts

- **Highest Checkout (Over/Under):** Modeled on exponential decay.
- **180s Match Bet:** A comparison of two independent Poisson rates.

2.9. Futsal

- **Half-Time/Full-Time (HT/FT):** Because futsal is so fast-paced, a team trailing at half-time has a statistically higher chance of mounting a comeback (turning a Loss/Win result) compared to traditional outdoor football.

3. Dynamic Market Parser (Handling Undefined Markets)

Bookmakers frequently invent new, obscure "novelty" or "special" markets (e.g., "Player X to hit the crossbar" or "Team A to win both halves and take 10+ corners").

The Odds Factory backend cannot have a hardcoded statistical model for everything. When the API receives an **undefined market**, it routes it to the **Dynamic Market Parser**.

How the Parser Works (Agnostic Optimization)

If the backend does not understand the *context* of the market, it relies entirely on mathematical arbitrage and **Wisdom of the Crowd**.

1. **Ingestion:** The engine detects an unmapped `market_id`.
2. **Market Matching:** It pings 5 to 10 top-tier global bookmakers (Pinnacle, Bet365, Betfair Exchange) to find the exact same string/market.
3. **Agnostic Devigging:** It extracts the Yes/No odds across the market.
 - *Example:* Undefined market implies 2.10 (Yes) / 1.70 (No).
 - *Implied probabilities:* Yes = 47.6%, No = 58.8%. Total = 106.4% (6.4% bookmaker margin).
4. **True Probability Extraction:** The engine applies Shin's Method to strip the 6.4% margin, revealing the market's true mathematical probability (e.g., Yes = 44.5%, No = 55.5%).
5. **Optimization/Editing:** Odds Factory can now allow the user to Edit, Cashout, or Optimize this bet using the True Probability (44.5%) without actually knowing anything about the sport or the players involved.

4. Continuous Learning Engine (AI Feedback Loop)

To maintain an edge, the Odds Factory statistical models (Poisson λ , Markov p/q) cannot be static. They must be updated by a Machine Learning engine.

The Bayesian Updating Loop

1. **Prior State:** The engine holds a "Prior" belief. (e.g., Manchester City's expected goals $\lambda = 2.4$).
2. **Observation:** The match is played. Man City scores 1 goal.
3. **CLV Measurement (Closing Line Value):** The AI checks the sharpest bookmakers directly before kickoff. If the sharp market shifted Man City's λ to 2.1 before the game started, Odds Factory logs an "Error Rate" in its own model.
4. **Posterior Update:** Utilizing Bayesian Inference, the AI updates its λ for the next match. It weights recent data heavier than older data (Exponential Moving Average).
5. **Player-Level Adjustments:** If a key player (e.g., a starting point guard in Basketball) is injured, the Continuous Learning system automatically detects the sudden shift in the global betting markets and retroactively adjusts the team's Offensive Efficiency rating downward before the next game.

Reinforcement Learning for Cashout Edits

The AI monitors user behavior during the "Edit Bet" phase.

- If the backend offers a cashout value of \$50, and 90% of users accept it, the AI learns the liquidity offer is attractive.
- If users continuously edit bets to remove specific markets (e.g., Users always remove "Player Props" when games go to halftime), the engine optimizes the UI and pricing algorithms to pre-calculate these specific swap routes faster, saving server compute overhead.

5. Summary of Data Flow for Unknown Markets

When a user clicks "Edit Bet" containing an unknown market: User Request -> Identify Market -> IF Known: Route to Sport Model (e.g., Poisson) -> ELSE: Route to Dynamic Parser -> Fetch Global Odds -> Strip Vigorish -> Extract True Probability -> Calculate Cashout -> Output to User.